Computationally Efficient Predictions for Multi-Output Spatiotemporal Dynamical Systems

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Abstract

Over the past decade, several groups of machine learning researchers have demonstrated promising computational algorithms and mathematical results which utilize Gaussian process (GP) regression and Kalman filtering (KF) to derive estimates and predictions for the future states of spatiotemporal dynamical systems in a more computationally-efficient manner.

This paper builds on previous results in this research area by adapting and extending the results of previous work on univariate outputs towards dynamical systems with vector-valued outputs. In this paper, we introduce the mathematical background necessary to understand Gaussian processes and Kalman filtering, and cover prior results in these areas. We then formally describe the multi-output spatiotemporal dynamics problem, and depict the assumptions required for similar computational speedups to exist in the multi-output spatiotemporal dynamics setting. Lastly, we apply these algorithms to both synthetic and real-world datasets to evaluate their performance.

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1 Introduction

Gaussian processes (GPs) are an 'industry standard' procedure for modelling the values of unknown functions, especially in scenarios where there are noisy or otherwise unreliable measurements and/or correlations in the data which must be accounted for (Rasmussen and Williams, 2006).

Gaussian process regression, also commonly referred to in the literature as *kriging* after the South African geostatistician Danie G. Krige (Cressie, 1990), is a statistical modelling procedure wherein a finite collection of observations are used for learning the form of an unknown real-valued function. Gaussian processes are particularly useful when there are correlations among observations, as these models can leverage information from correlated data to reduce the variance of outputs in the estimated function. Furthermore, the equations for fitting a GP to a dataset have computable closed-form solutions. Due to these convenient properties, Gaussian processes are frequently used in a variety of practical statistical applications involving spatiotemporal datasets, ranging from geostatistics and climatology to fields such as electrical engineering and signal processing (Särkkä and Hartikainen, 2012; Huang et al., 2025).

Although Gaussian processes are mathematically tractable and convenient to use, there is one major drawback to these models: exact inference with Gaussian processes has a cubically-scaling computational overhead. Fitting a Gaussian process on a set of N observations is a process which requires $\mathcal{O}(N^3)$ operations (Rasmussen and Williams, 2006), making these regression problems infeasible for many large-scale datasets (Guhaniyogi and Banerjee, 2018; Pförtner et al., 2025; Salari and Zeinali, 2025).

Due to these computational difficulties, many statisticians and machine learning researchers have sought to develop algorithms for fitting a Gaussian process with reduced computational overhead. Prominent well-studied and broadly applicable methods for accomplishing this task include approximations using induced latent variables (Quiñonero-Candela and Rasmussen, 2005), sparse variational Gaussian processes (Titsias, 2009), and Vecchia approximations (Katzfuss and Guinness, 2021). However, these methods all require some degree of approximation, trading the exactness of a solution for increased algorithmic efficiency.

In their 2020 paper *Efficient spatio-temporal Gaussian regression via Kalman filtering*, Todescato et al. introduce a new algorithm for efficiently deriving estimates of the future states in a spatio-temporal dynamical system under certain assumptions for the covariance structure between observations. The algorithm proposed by the authors utilizes Gaussian process regression in tandem with the Kalman filter, providing the ability to account for both the spatial and temporal aspects which guide the 'evolution' of the dynamical system while simultaneously reducing the overall computational load.

This project gives a detailed overview and subsequently builds upon the results in Todescato et al. (2020) by extending their proposed algorithm to operate with vector-valued outputs instead of being confined to scalar responses. This alteration provides many practical benefits, as there are many scientific research areas which utilize spatiotemporal models for multiple correlated responses.

2 Background

In order to properly describe the algorithm devised by Todescato et al. (2020), we must first separately discuss Gaussian processes and Kalman filtering.

2.1 Gaussian Processes

A Gaussian process (GP) is a real-valued stochastic process indexed over some P-dimensional *state space* $\mathcal{X} \subseteq \mathbb{R}^P$ which satisfies the following key property:

Definition 1 (Gaussian Process)**.** *The real-valued function $f : \mathcal{X} \to \mathbb{R}$ follows a Gaussian process on $\mathcal{X}$ if for every finite collection (N-tuple) of states $\{\mathbf{x}_1, \cdots, \mathbf{x}_N\} \subset \mathcal{X}$, the joint distribution of the random vector $\big(f(\mathbf{x}_1), \cdots, f(\mathbf{x}_N)\big)^\mathsf{T}$ is a multivariate Gaussian distribution on $\mathbb{R}^N$.*

Furthermore, every Gaussian process on the state space $\mathcal{X}$ is *uniquely* specified by two real-valued functions: a mean function $m : \mathcal{X} \to \mathbb{R}$ and a *positive-definite* covariance (kernel) function $k : \mathcal{X} \times \mathcal{X} \to \mathbb{R}^{\geq 0}$, which respectively describe the mean and covariance of the underlying process (Rasmussen and Williams, 2006). Any function $f : \mathcal{X} \to \mathbb{R}$ which is a realization of a Gaussian process with mean $m(\cdot)$ and kernel $k(\cdot, \cdot)$ is endowed with the following structure:

$$\textbf{Mean:} \quad m(\mathbf{x}) = \mathbb{E}[f(\mathbf{x})] \text{ for all } \mathbf{x} \in \mathcal{X} \tag{1a}$$

$$\textbf{Covariance:} \quad k(\mathbf{x}, \mathbf{x}') = \mathrm{Cov}(f(\mathbf{x}), f(\mathbf{x}')) \text{ for all } (\mathbf{x}, \mathbf{x}') \in \mathcal{X} \times \mathcal{X} \tag{1b}$$

We denote a function $f : \mathcal{X} \to \mathbb{R}$ which is drawn from a Gaussian process with mean $m(\cdot)$ and kernel $k(\cdot, \cdot)$ as $f \sim \mathcal{GP}(m, k)$. Note that the kernel function $k(\cdot, \cdot)$ in Equation (1b) must be symmetric in its arguments, as this function describes the covariance between the Gaussian process outputs at these two locations. In a Bayesian setting, the structure of a Gaussian process makes these models highly suitable for use as a prior distribution on a space of real-valued functions (Titsias, 2009; Álvarez et al., 2012), as the Gaussian distribution is a conjugate prior which yields a Gaussian posterior distribution.

Suppose that $f \sim \mathcal{GP}(m, k)$ is a realization of some Gaussian process on $\mathcal{X} \subseteq \mathbb{R}^P$, and let $\{\mathbf{x}_1, \ldots, \mathbf{x}_N\} \subset \mathcal{X}$ be a set of queried locations. Define the data matrix $\mathbf{X} := \big[\, \mathbf{x}_1^\mathsf{T} \cdots \mathbf{x}_N^\mathsf{T} \,\big]^\mathsf{T} \in \mathbb{R}^{N \times P}$, and define the corresponding vectors $f(\mathbf{X}) := \big[\, f(\mathbf{x}_1^\mathsf{T}) \cdots f(\mathbf{x}_N^\mathsf{T}) \,\big]^\mathsf{T} \in \mathbb{R}^N$ and $m(\mathbf{X}) := \big[\, m(\mathbf{x}_1^\mathsf{T}) \cdots m(\mathbf{x}_N^\mathsf{T}) \,\big]^\mathsf{T} \in \mathbb{R}^N$. Lastly, let $k(\mathbf{X}, \mathbf{X}) \in \mathbb{R}^{N \times N}$ represent the *Gram matrix* induced by the kernel $k(\cdot, \cdot)$:

$$k(\mathbf{X}, \mathbf{X}) := \begin{bmatrix} k(\mathbf{x}_1, \mathbf{x}_1) & k(\mathbf{x}_1, \mathbf{x}_2) & \cdots & k(\mathbf{x}_1, \mathbf{x}_n) \\ k(\mathbf{x}_2, \mathbf{x}_1) & k(\mathbf{x}_2, \mathbf{x}_2) & \cdots & k(\mathbf{x}_2, \mathbf{x}_n) \\ \vdots & \vdots & \ddots & \vdots \\ k(\mathbf{x}_n, \mathbf{x}_1) & k(\mathbf{x}_n, \mathbf{x}_2) & \cdots & k(\mathbf{x}_n, \mathbf{x}_n) \end{bmatrix} \tag{2}$$

One of the key properties which allows Gaussian processes to be so widely-applicable is that under the conditions described above, the vector $f(\mathbf{X})$ follows a multivariate Gaussian distribution:

$$f(\mathbf{X}) \sim \mathcal{N}\Big(m(\mathbf{X}), k(\mathbf{X}, \mathbf{X})\Big) \tag{3}$$

Furthermore, consider the same setting as above, and suppose that we have some (potentially) noisy measured observations given by $y_i = f(\mathbf{x}_i) + \varepsilon_i$, for independent normally-distributed mea-

surement errors $\varepsilon_1, \ldots, \varepsilon_N \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2)$. Equivalently, $y_i \sim \mathcal{N}\big(f(\mathbf{x}_i), \sigma^2\big)$ for each queried location $\mathbf{x}_i$ (Rasmussen and Williams, 2006). Defining the response vector $\mathbf{y} := \begin{bmatrix} y_1 & \cdots & y_N \end{bmatrix}^\mathsf{T} \in \mathbb{R}^N$ yields the following distributional result:

$$\mathbf{y} \sim \mathcal{N}\Big(m(\mathbf{X}), k(\mathbf{X}, \mathbf{X}) + \sigma^2 \mathbf{I}\Big) \tag{4}$$

In a Bayesian context, one can use a Gaussian process as a prior on the space of real-valued functions on $\mathcal{X}$. If observations are measured with normally-distributed measurement error, then the posterior predictive distribution for the function can be described by a Gaussian process with a similar, but altered, mean and kernel. In particular, for a data matrix $\mathbf{X} \in \mathbb{R}^{N \times P}$ with the response vector $\mathbf{y} \in \mathbb{R}^N$, the posterior predictive distribution for f evaluated at any points $\mathbf{x}, \mathbf{x}' \in \mathcal{X}$ is given by the following:

$$\textbf{Mean:} \quad m_{\text{post}}(\mathbf{x}) = m(\mathbf{x}) + k(\mathbf{x}, \mathbf{X})\big[k(\mathbf{X}, \mathbf{X}) + \sigma^2 \mathbf{I}\big]^{-1}\big(\mathbf{y} - m(\mathbf{X})\big) \tag{5a}$$

$$\textbf{Kernel:} \quad k_{\text{post}}(\mathbf{x}, \mathbf{x}') = k(\mathbf{x}, \mathbf{x}') - k(\mathbf{x}, \mathbf{X})\big[k(\mathbf{X}, \mathbf{X}) + \sigma^2 \mathbf{I}\big]^{-1} k(\mathbf{X}, \mathbf{x}') \tag{5b}$$

Note that if the observations $y_1, \ldots, y_N$ are known to be exact (i.e. $\sigma^2 = 0$), then Equations (5a) and (5b) will perfectly interpolate every observation. To be precise, for any queried point $\mathbf{x}_i$, we will have that $m_{\text{post}}(\mathbf{x}_i) = y_i$ and $k_{\text{post}}(\mathbf{x}_i, \mathbf{x}_i) = 0$, implying that the posterior is a deterministic value at that location. This property holds if and only if σ^2 is identically zero.

2.2 Multi-Output Gaussian Processes

Thus far, we have focused solely on the description of single-output Gaussian processes, whose realizations are described by functions of the form $f : \mathcal{X} \to \mathbb{R}$. However, there is a natural extension from these univariate processes to *multi-output* (also called *multi-task*) Gaussian processes: a higher-dimensional form of a Gaussian process, whose outputs are vectors in $\mathbb{R}^D$ (Álvarez et al., 2012; Bonilla et al., 2007).

2.3 Kalman Filtering

The Kalman filter (KF), first introduced by and eponymously named for Rudolf E. Kálmán (1960), is a technique for

3 Problem Formulation

In

Definition 2 (Power Spectrum Density)**.**

4 Methods

4.1 Code Implementation

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5 Results

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6.1 Conclusions

6.2 Limitations

6.3 Future Work

References

Álvarez, M. A., Rosasco, L., and Lawrence, N. D. (2012). Kernels for Vector-Valued Functions: A Review. *Foundations and Trends in Machine Learning*, 4(3):195–266.

Bonilla, E., Chai, K., and Williams, C. (2007). Multi-task Gaussian Process Prediction. In Platt, J., Koller, D., Singer, Y., and Roweis, S., editors, *Advances in Neural Information Processing Systems*, volume 20. Curran Associates, Inc.

Cressie, N. (1990). The origins of kriging. *Mathematical Geology*, 22(3):239 – 252.

Guhaniyogi, R. and Banerjee, S. (2018). Meta-Kriging: Scalable Bayesian Modeling and Inference for Massive Spatial Datasets. *Technometrics*, 60(4):430–444. PMID: 31007296.

Huang, H., Blake, L. R., Katzfuss, M., and Hammerling, D. M. (2025). Nonstationary Spatial Modeling of Massive Global Satellite Data. *Journal of Computational and Graphical Statistics*, 34(4):1226–1239.

Kálmán, R. E. (1960). A New Approach to Linear Filtering and Prediction Problems. *Journal of Basic Engineering*, 82(1):35–45.

Katzfuss, M. and Guinness, J. (2021). A General Framework for Vecchia Approximations of Gaussian Processes. *Statistical Science*, 36(1):124 – 141.

Pförtner, M., Wenger, J., Cockayne, J., and Hennig, P. (2025). Computation-Aware Kalman Filtering and Smoothing. In Li, Y., Mandt, S., Agrawal, S., and Khan, E., editors, *Proceedings of The 28th International Conference on Artificial Intelligence and Statistics*, volume 258 of *Proceedings of Machine Learning Research*, pages 2071–2079. PMLR.

Quiñonero-Candela, J. and Rasmussen, C. E. (2005). A Unifying View of Sparse Approximate Gaussian Process Regression. *Journal of Machine Learning Research*, 6(65):1939–1959.

Rasmussen, C. E. and Williams, C. K. I. (2006). *Gaussian Processes for Machine Learning*. The MIT Press.

Salari, A. and Zeinali, M. (2025). Kalman-Enhanced Dynamic Spatiotemporal Gaussian Process Regression for Real-Time Wind Turbine Power Estimation. In *2025 15th Smart Grid Conference (SGC)*, pages 1–4.

Särkkä, S. and Hartikainen, J. (2012). Infinite-Dimensional Kalman Filtering Approach to Spatio-Temporal Gaussian Process Regression. In Lawrence, N. D. and Girolami, M., editors, *Proceedings of the Fifteenth International Conference on Artificial Intelligence and Statistics*, volume 22 of *Proceedings of Machine Learning Research*, pages 993–1001, La Palma, Canary Islands. PMLR.

Särkkä, S. and Svensson, L. (2023). *Bayesian Filtering and Smoothing*. Institute of Mathematical Statistics Textbooks. Cambridge University Press, 2 edition.

Titsias, M. K. (2009). Variational Learning of Inducing Variables in Sparse Gaussian Processes. In van Dyk, D. and Welling, M., editors, *Proceedings of the Twelfth International Conference on Artificial Intelligence and Statistics*, volume 5 of *Proceedings of Machine Learning Research*, pages 567–574, Hilton Clearwater Beach Resort, Clearwater Beach, Florida USA. PMLR.

Todescato, M., Carron, A., Carli, R., Pillonetto, G., and Schenato, L. (2020). Efficient spatio-temporal Gaussian regression via Kalman filtering. *Automatica*, 118:109032.

Appendix

A Definitions and Technical Details

B Code and Simulation Outputs

For the sake of brevity, the code used in this project is not directly added to the appendix. All code is available in a public GitHub repository, which can be located at https://github.com/hotz1/Dynamical-Multitask-Prediction.